

Bachelor Project



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Technical
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F3

**Faculty of Electrical Engineering
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Design of a platform for 2D LiDaR

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Field of study: Electronics and Communications

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Acknowledgements

Děkuji ČVUT, že mi je tak dobrou *alma mater*.

Declaration

Prohlašuji, že jsem předloženou práci vypracoval samostatně, a že jsem uvedl veškerou použitou literaturu.

V Praze, 13. November 2025

Abstract

There are currently many advanced LiDAR sensors available for 3D spatial scanning [*insert citation]. However, most of these sensors are too expensive for the average user. The most affordable commercial option is the Unitree L1, which costs \$ 250 [*insert citation]. The aim of this miniproject is to design a platform for a 360° LiDAR sensor, which will be utilized in a subsequent bachelor's thesis to visualize a 3D image of the scanned environment.

Keywords: LiDAR, 3D reconstruction, point cloud, signal processing, mapping

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Abstrakt

V současnosti existuje spousta pokročilých lidar-senzorů určených pro 3D snímání prostoru [*insert citation*]. Většina z nich je však cenově nedostupná pro běžného uživatele. Nejvíce dostupným komerčním řešením je Unitree L1, který stojí \$ 250 [*insert citation*]. Cílem toho miniprojektu je navrhnout pro 360° lidar-senzor platformu a použít ji v navazující bakalářské práci pro vizualizaci 3D-obrazu naskenovaného prostoru.

Klíčová slova: LiDAR, 3D rekonstrukce, point cloud, zpracování signálů, mapování

Překlad názvu: Návrh platformy pro 2D LiDaR

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Acronyms

LiDAR Light Detection and Ranging

DSP Digital Signal Processing

SLAM Simultaneous Localization and Mapping



Chapter 1

Introduction

Nowadays, LiDAR sensors are widely used in various applications, including autonomous vehicles, robotics, and environmental mapping [insert citation for robotics BAP and Pinliang book]. While the former two applications often require high-end and expensive LiDAR systems, environmental mapping can be effectively performed using more affordable sensors. However, even the most affordable commercial LiDAR sensors which was found during the research for this miniproject cost approximately \$250[insert Unitree citation]. The primary motivation behind this miniproject (and the subsequent bachelor thesis) is to develop a cost-effective platform for a 360° LiDAR sensor that can be utilized for 3D spatial scanning and visualization of the surrounding environment. A subsequent bachelor thesis will then focus on the Digital Signal Processing (DSP) techniques for effective processing of the acquired point cloud data and proper utilization of open-source visualization tools (such as Python's Open3D library[insert citation of a site or a thesis for that library]) to recreate a 3D scene.

Chapter 2

Design of a Platform

This chapter will firstly go over the existing DIY solutions for 360° LiDAR platforms which the author has stumbled upon during the research[too informal?], analyzing their pros and cons.

2.1 Existing DIY Solutions for Environmental Mapping

2.1.1 Using a Single-Point LiDAR module

One of the most straightforward approaches to achieve 360° environmental mapping is to utilize a single-point LiDAR module mounted on a rotating platform, as was done in this project[add citation and picture].

Pros:

- Single-point LiDARs are the most cost-effective among all existing types
- Simple design

Cons:

- Slow data acquisition relative to methods leveraging 2D LiDAR's speed
- Movement of lidar may introduce inaccuracies in distance measurements, which need to be compensated for in code if we want to assume that all points are measured from a single position in space

■ 2.1.2 Using a Single-Point LiDAR and a Mirror

The following approach was used in this project[add citation and picture]. A single-point LiDAR is placed below a rotating mirror, which reflects the laser beam horizontally. As the mirror rotates, the laser beam scans the environment in 360°. Moreover, by tilting the mirror at an angle, vertical scanning can also be achieved, allowing for the acquisition of the whole scene.

Pros:

- Cost-effectiveness as in previous implementation
- Simple design
- Much easier to correct the distance measurement, since the LiDAR is stationary and the distance from it to the mirror is known.

Cons:

- Slow data acquisition relative to methods leveraging 2D LiDAR's speed
- Mirror's quality may affect the accuracy of distance measurements
- Mechanical structure holding the mirror may introduce vibrations, thus affecting measurement accuracy

■ 2.1.3 Using a 2D LiDAR Mounted on a Tilting Platform

This approach utilizes a 2D LiDAR sensor mounted on a platform that can tilt the sensor side-to-side, as demonstrated in this project[add citation and picture]. The 2D LiDAR scans the environment in a horizontal plane, while the tilting mechanism allows for vertical scanning. Pros:

- Faster data acquisition compared to solutions with single-point LiDAR measurements
- Higher resolution given by the use of a 2D LiDAR sensor
- This design could be potentially used for Simultaneous Localization and Mapping (SLAM) applications, given the higher speed

Cons:

- More expensive than single-point LiDAR solutions
- Point distribution is non-uniform due to the tilting mechanism which produces denser point clouds on the axis of tilting (see MATLAB simulation below) [should this note be here?]
- Mechanical design is both more complex and more prone to vibrations produced by the LiDAR itself affecting measurement accuracy

2.2 Proposed Design of a 360° LiDAR Platform

Based on the analysis of existing DIY solutions for 360° LiDAR platforms, the proposed design for this miniproject involves using a 2D LiDAR sensor mounted on a tilted platform that is able to rotate 360°. This design leverages the advantages of 2D LiDAR sensors, such as fast data acquisition and high resolution, while also eliminating some of the drawbacks of previous designs, namely:

- The platform will be quite rigid, minimizing vibrations
- Mechanical design is very simple, making it easy to build and troubleshoot
- The point cloud will be more uniformly distributed compared to the tilting platform design, as the rotation will ensure that a LiDAR won't have a preferred direction around which it collects denser data unless the slope is very steep (see MATLAB simulation below)[could this be phrased better?]

The main drawback of this design is the existence of "blind spots" in the point cloud, which are areas that the LiDAR cannot scan due to the tilt angle.

However, this issue can be mitigated by carefully selecting the tilt angle based on the specific application in which the system will be used. (more on that in the next chapter) [system? :/]



Chapter 3

Conclusion